

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460694.6441 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.017
Transit depth (Rp/Rs)^2	2.48 +/- 0.53 [%]
Orbital Inclination (inc)	83.43 +/- 0.69
Airmass coefficient 1 (a1)	0.9543 +/- 0.0084
Airmass coefficient 2 (a2)	0.0312 +/- 0.0076
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	#2 - [311, 221]
Optimal Aperture	3.89
Optimal Annulus	9.86
Transit Duration (day)	0.0643 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.017 vs 0.1404 (deviation 0.98σ)
Duration fit vs published	0.0643 d vs 0.0754 d (deviation 1.29σ)
Mid-time O–C	3.4 min
Depth SNR	9.2
Residual scatter	0.88 %

Light curve

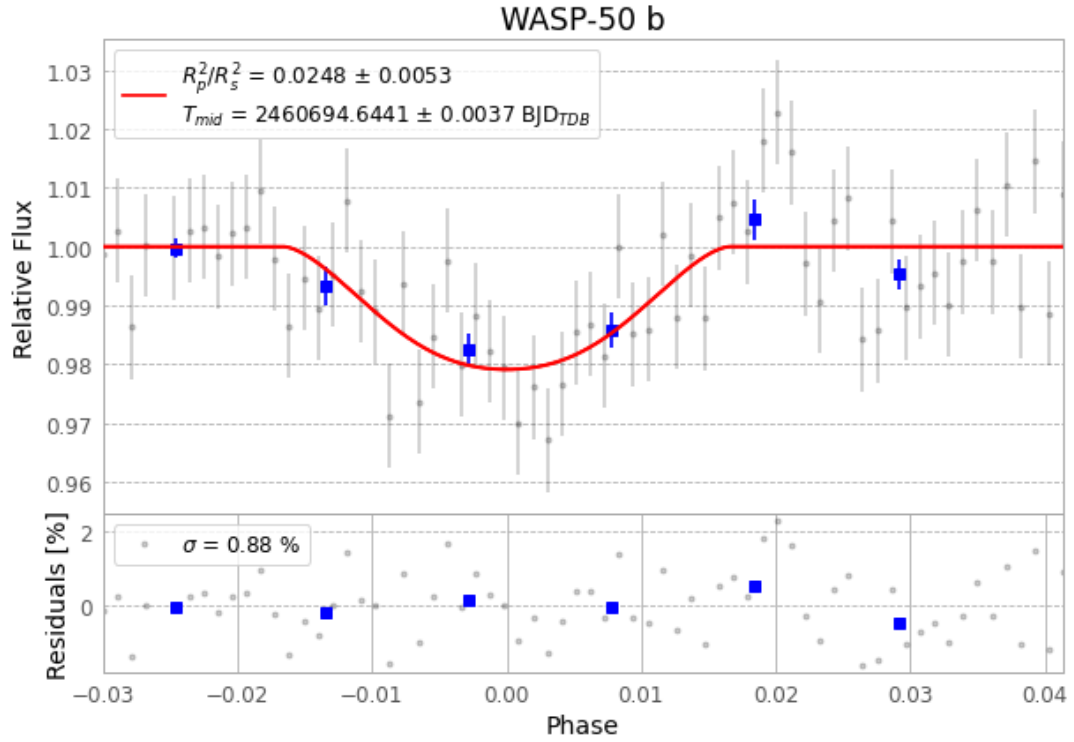


Figure 1 — FinalLightCurve_WASP-50 b_2025-01-18.png (SHA-256 2b0253a27fc9297e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [323, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 722312709678ec49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

68 FITS frames (45 MB), WASP-50250119020016.FITS → WASP-50250119052115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-250119.log (e0d0b29cecab...), FinalLightCurve_WASP-50 b_2025-01-18.png (2b0253a27fc9...), FinalLightCurve_WASP-50 b_2025-01-18.csv (8fd3deac4e7c...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:31Z.